EXTRACTION OF DIFFERENT GARLIC VARIETIES (*Allium sativum* L.) – DETERMINATION OF ORGANOSULFUR COMPOUNDS AND MICROBIOLOGICAL ACTIVITY

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ABSTRACT

Water and ethanol garlic extracts were prepared for encapsulation into double emulsion carriers. For extracts preparation three varieties of garlic were used: NSPBL-70, *Labud* and *Bosut*. Determination of organosulfur compounds was done by HPLC-DAD method. Results indicated the differences in sulphur compounds concentrations depending on garlic variety and solvent used for extraction. Dominant sulfur compound in water garlic extract was allicin (AC), with concentrations from 42.74 to 50.79 µg/ml, depending on garlic variety, where extract of *Labud* had the highest AC concentration. Other investigated sulfur compounds were allyl sulfide (AS), diallyl disulfide (DADS) and methanethiosulfonic acid *S*-methyl ester (MMTS). AC was also dominant sulfur compounds in ethanol garlic extract, but in significantly lower concentrations than in water garlic extract (4.39 - 4.56 µg/ml). Extraction with ethanol showed small differences in sulfur compounds composition between extracts obtained from different varieties of garlic. AC content in lyophilized aqueous garlic extract was 12.35 ng/mg, while lyophilized ethanol garlic extract had about ten times smaller concentration of AC (1.05 ng/mg), but higher concentration of MMTS. Antibacterial activity of garlic extracts on growth of *Staphylococcus aureus* and *Escherichia coli* were determined using micro-dilution method. Microbiological activity of extracts was dependent on the garlic variety and extraction solvent. *Labud* water garlic extract had shown the smallest minimum inhibitory concentration, for both examined bacteria strains, which may be linked with highest AC concentration. The effect of ethanol garlic extract on the growth of *E. coli* does not depend on the variety of garlic, which is in accordance with small differences in the sulfur compounds composition, while for growth inhibition of *S. aureus*, the best result showed NSPBL-70 ethanol extract.
Keywords: *garlic extract, organosulfur compounds, allicin, antibacterial activity*

INTRODUCTION

The garlic (*Allium sativum* L.) is considered as a nutrient with various positive effect on human health, which is confirmed trough many scientific studies (Das et al., 2012; Suleria et al., 2015). Consumption of the garlic and its products is associated with the cold and flu prevention, prevention and treatment of cardiovascular diseases, hypertension, thrombosis, atherosclerosis, hyperlipidemia, and diabetes (Hosseini and Hosseinzadeh, 2015; Majewski, 2014). The other potential biological effects are associated with the reduction of risk factors for cancer, stimulation of immune function, improved detoxification, hepatitis prevention and antioxidant properties (Nicastro et al., 2015; Percival, 2014). In addition, the garlic was characterised as a plant with very good antimicrobial, antifungal, antiparasitic and antiviral activity (El-Sayed et al., 2017; Majewski, 2014; Suleria et al., 2015).
The most active ingredients of garlic are volatile components (thiosulfonates and other organosulfur components), which are responsible for the characteristic smell and the taste of garlic itself. Tiosulfonates are reactive and unstable compounds, susceptible to spontaneous chemical reactions leading to a wide variety of sulfur compounds that take part in further transformations, without losing its biological activity (Ramirez et al., 2017). However, due to their thermal instability, the main biologically active tiosulfonates are destroyed in the process of food preparation (Lanzotti, 2006).

One of the most important ingredients of the garlic is allicin (AC). It is considered to be responsible for antibacterial activity, and many other biological effects such as antiatherosclerotic, antithrombotic, antiinflammatory, immunomodulatory and anti-cancer effects (Harris et al., 2001; Hussein et al., 2017). Among other active compounds, it was reported that diallyl trisulfide (DATS) and diallyl disulfide (DADS) are the most active against yeasts, while ajoene is the main compound responsible for the antiviral activity. Allicin and propyl propane-TS (PPTS) have the strongest antiplatelet activity (thrombosis prevention), while ethyl ethane-TS (EETS) and methyl methane-TS (MMTS) are much weaker, but are still significantly more potent platelet inhibitors than aspirin at nearly equivalent concentrations (Briggs et al., 2000). Although allicin has been reported as a main anticancer compound, other garlic sulfur compounds (allyl sulfide, allyl disulfide, diallyl disulfide, allyl trisulfide, and SAMC) have been reported to inhibit proliferation of tumour cells (Lee et al., 2013).

A mass consumption of the garlic is mostly restricted by its distinctive smell and taste. Encapsulation of biologically active substances and active ingredients into water-in-oil (W/O) and water-in-oil-in-water (W/O/W) emulsions, can give possibility to achieve controlled release of active ingredients and flavor retention, mask the bad taste or smell of some components, stabilize food ingredients, prevent their oxidation or hydrolysis, adjust their properties and/or increase their bioavailability (Ilić et al., 2017). In this way, the encapsulation of the garlic extract into emulsion carriers may contribute to increased consumption.

In this research we explore the composition of different garlic extracts (produced from three garlic varieties), and aim to choose the best one for encapsulation into double emulsion carriers, in terms of the highest level of investigated organosulfur compounds and the highest antimicrobial activity. In particular, this paper will present comparison between organosulfur compounds concentrations of lyophilisedgarlic extracts prepared with different extraction procedures.

MATERIAL AND METHODS

Plant materials used for extraction were fresh garlic gloves (*A. sativum* L.*)* of three different garlic varieties: NSPBL-70, *Labud* and *Bosut*. All garlic varieties were provided by Inst. of Field and Vegetable Crops (Novi Sad, Serbia), cultivated on their experimental fields, while NSPBL-70 and *Bosut*are accepted varieties in Serbia.Demineralised water and 96% ethanol (Zorka Pharma d.o.o (Šabac, Serbia)) were used for extraction.

Preparation of water garlic extract

Fresh garlic bulbs (90g) were mixed with demineralised water (200g) and grinded in a blender for 15 min. The solid parts of the garlic were removed by filtration through sterile gauze, followed by centrifugation at 4500 rpm for 30 min at 20 °C. The supernatant was filtrated through filter paper using vacuum pump (Heidolph Instruments GMbH, Germany). Finally, the extract was collected and stored in refrigerator at 4°C (Ilić et al., 2017). The extract samples were lyophilised for 3 days.

Preparation of ethanol garlic extract

The recipe for the preparation of ethanol extract used in this work originates from ancient times, where it was used as a folk remedy. Nowadays, the composition based on this recipe is commercially available in the form of drops (BioCapilary kapi, Priroda na dar). For the preparation, fresh garlic bulbs (350g) were chopped into small pieces, mixed with 96% ethanol (300g) and left for 10 days in the dark glass bottle protected from the light at room temperature, with occasional mixing. The solids parts of the garlic were removed by filtration through sterile gauze, centrifuged at 4500 rpm for 30 min at 20 °C and finally filtrated under vacuum (Heidolph Instruments GMbH, Germany). Extract was stored in a dark bottle. Before same procedure lyophilisation process employed for water garlic extracts, samples of ethanol garlic extracts were evaporated at 45 °C.

Chemical analysis

The determination of the organosulfur compounds was done by HPLC-DAD analysis, on a Poroshell 120 EC-C18 (100×3 mm, particles 2.7 µm) HPLC column. The column oven was conditioned at 30°C. Linear gradient elution mode with the mobile phase flow rate of 0.8 ml/min was used. The initial soluble A ratio (phosphate buffer 20 mM, pH 4.5) and solvent B (acetonitrile) was 95% B; 0-5 min, 5% B; 5-15 min, 5-50% % B; 15-20 min, 50% B; 20-22 min, 50-80% B; 22-25 min, 80% B; 25-30 min, 80-95% B. An automatic injector was used to subject samples of a 5 µl volume into the instrument. The analysis time was 30 minutes, while the column washout time was 3 minutes (Tomsik et al., 2018 under review). The spectrum is set for shooting in the range of 190-400 nm, while the chromatogram is recorded at 210 and 240 nm. Compounds such as allicin (AC), allyl sulfide (AS), dialyl disulfide (DADS), and S-methyl ester methanesulfonic acid (MMTS) have been identified based on the retention time and spectral characteristics of the standard.

The liquid extracts were diluted with acetonitrile (ACN) filtered through a 0.45 µm nylon filter (Rotilabo-Spritzenfilter 13 mm, Roth, Karlsruhe, Germany). The lyophilized extract was reconstituted in a suitable solvent and sonicated for 30 min, diluted with ACN and finally filtered through a 0.45 µm nylon filter.

Microbiological analysis

Minimal inhibitory concentration (MIC) and minimal bactericidal concentration (MBC) were obtained using broth microdilution method. The assay was carried out using the following bacterial strains: Gram positive *Staphylococcus aureus* (ATCC 25923) and gram negative *Escherichia coli* (ATCC 25922).

RESULTS AND DISCUSSION

The investigation of extracts prepared from different garlic varieties was performed in order to select the garlic variety with the highest level of organosulfur compound concentrations, for its further encapsulation in emulsion carriers. Table 1 summarizes AC, DADS, AS and MMTS concentrations in water and ethanol garlic extracts prepared from garlic varieties NSPBL-70, *Labud* and *Bosut*.

Table 1. Organosulfur compound concentrations in water and ethanol garlic extracts prepared from different garlic varieties

Garlic variety	**Organosulfur compound**			
	AC (µg/ml)	**DADS (µg/ml)**	**AS (µg/ml)**	**MMTS (µg/ml)**
	Water garlic extract (WatGE)			
NSPBL-70	42.74 ± 0.52	0.01 ± 0.00	2.89 ± 0.06	0.09 ± 0.01
Labud	50.79 ± 0.83	0.02 ± 0.00	4.72 ± 0.08	0.25 ± 0.01
Bosut	45.93 ± 0.65	0.03 ± 0.00	1.91 ± 0.02	0.33 ± 0.02
	Ethanol garlic extract (EthGE)			
NSPBL-70	4.56 ± 0.19	0.04 ± 0.00	0.24 ± 0.01	0.45 ± 0.02
Labud	4.41 ± 0.12	0.03 ± 0.00	0.21 ± 0.01	0.67 ± 0.03
Bosut	4.39 ± 0.23	0.03 ± 0.00	0.70 ± 0.02	0.56 ± 0.01

As can be seen from Table 1, the sulfur compounds concentrations are depend, not only on garlic variety, but also on procedure used for extraction. Allicin was dominant compound in all three investigated varieties, with concentrations between 42.74 and 50.79 µg/ml. Water garlic extract obtained by extraction of Labud had the highest AC (50.79 ± 0.23 µg/ml) and AS (4.72 ± 0.08 µg/ml) concentrations. Bosut water extract showed the highest concentration of DADS, in comparison with other investigated varieties. NSPBL-70 water extract had the smallest AC, DADS and MMTS concentrations. The change of solvent and the extraction procedure gave extracts with much lower concentrations of allicin. Namely, ethanol garlic extracts had about ten times lower concentrations of AC, than water garlic extracts, which

was in the range between 4.39 and 4.56 µg/ml. The variation of AC concentration was minor among different garlic varieties. Also, the content of other investigated components in ethanol garlic extracts did not change significantly with the garlic variety.
A comparison between lyophilized extracts obtained using different extraction procedures have shown some differences in concentration of investigated sulfur compounds. Results are shown in Table 2. Concentration of AC in lyophilized ethanol garlic extract was 1.05 ± 0.08 ng/mg, ten times smaller than concentration in water extract (12.35 ± 0.06 ng/mg). Ethanol garlic extract had somewhat higher concentration of DADS and MMTS in comparison with extract obtained by extraction with water, while AS content was similar in both investigated samples.

Table 2. Organosulfur compound concentrations in lyophilized water and ethanol garlic extracts

	Organosulfur compound			
	AC (ng/mg)	**DADS (ng/mg)**	**AS (ng/mg)**	**MMTS (ng/mg)**
Water garlic extract	12.35 ± 0.06	0.16 ± 0.01	1.31 ± 0.06	1.42 ± 0.07
Ethanol garlic extract	1.05 ± 0.08	0.57 ± 0.03	0.99 ± 0.04	3.33 ± 0.15

The antibacterial activity of investigated extracts obtained from different garlic varieties is given in Fig.1. The influence of extracts on growth of *S. aureus* and *E.coli* was explored, and corresponding MICs and MBCs were obtained. As can be noticed, results of MICs and MBCs depend on the garlic variety, as well as procedure used for extraction.
Labud water garlic extract had shown the smallest MIC (56.82 µl/ml), for both examined bacteria strains, while the corresponding MBC was 227.27 µl/ml. The best antibacterial activity of *Labud* water extract in comparison to other variety of garlic may be a consequence of the highest AC concentration (Table 1). Antibacterial properties of garlic is often connected to antimicrobial effect of AC, which was reported to exhibit antibacterial activity against a wide spectrum of bacteria (*Escherichia, Salmonella, Staphylococcus, Streptococcus, Klebsiella, Proteus, Bacillus, Clostridium, Mycobacterium tuberculosis and Helicobacter pylori*) and fungi (*Candida, Cryptococcus, Trichophyton, Epidermophyton, and Microsporum*) (Ankri & Mirelman, 1999). Water extract of NS*PBL*-70 also gave a low MIC value only for *S. aureus* (MIC = 56.82 µl/ml), while for the growth inhibition of *E. coli* a higher concentration of extract was required (MIC = 227.27 µl/ml and MBC = 454.54 µl/ml).
The effect of ethanol garlic extract on the growth of E. coli is not dependent on the variety of garlic, whereby the same MIC (113.64 µl/ml) and MBC (227.27 µl/ml) were obtained for all investigated extracts. This effect could be a consequence of a small difference of sulfur compounds composition in all investigated samples. However, the influence of ethanol garlic extracts on growth inhibition of S. aureus was somewhat different, where the best result showed NSPBL-70, with the same MIC and MBC as well as for E. coli. Labud or Bosut ethanol extract must be applied in higher concentrations, where for the growth inhibition 227.27 µl/ml of extractwas necessary, while appropriate MBCs were 454.54 µl/ml.

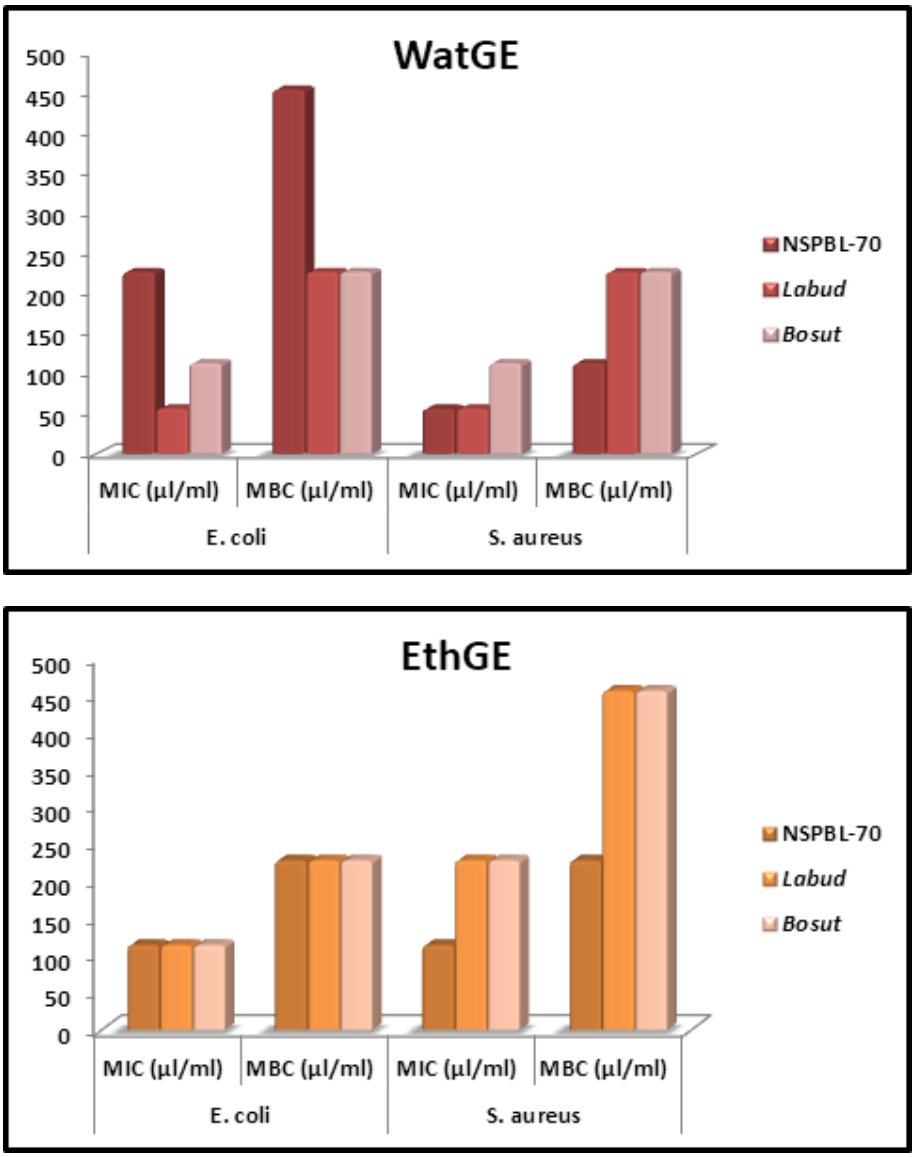

Figure 1. *Antibacterial activity of water and ethanol garlic extracts (WatGE and EthGE) prepared with different varieties of garlic*

CONCLUSIONS

The garlic extracts obtained from different garlic varieties, using two extraction solvents, were analysed in order to select the appropriate variety of garlic for encapsulation into double emulsion carriers. In this regard, the organosulfur compounds concentrations and antibacterial activity of extracts has been investigated. Results indicated that *Labud* water garlic extract has the best performance in terms of the highest AC content, the component that was often associated with positive influence on human health. The microbiological study also favors *Labud* water garlic extract, with the smallest MICs and MBCs necessary for growth inhibition of investigated bacteria. When ethanol was used for extraction, all investigated varieties of garlic gave similar sulfur compound concentrations and antibacterial activities.

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